PM Graded Quiz M2

Question 1

How many parameters do we need to estimate to perform a classical mean-variance optimization when we have 25 assets?

- 325
- 375
- 350
- 650

Question 2

What is the most sensible response to the following statement:

"Mean-variance optimization to determine the optimal portfolio is a fascinating tool, but it suffers from so many issues that it can hardly be expected to have any relevance for a practitioner."

- No big surprise! It is just another of those cases where there is a big discrepancy between theoretical merits and the ability of a methodology to be meaningful in real-life applications.
- While there are definitely issues, if we are very careful in its application and use some statistical sophistication, we can still produce meaningful results.
- While the criticism is deserved, since we do not have an alternative, it is the best we can do, and we should keep using it since something is better than nothing.
- It is too harsh! If we use the right statistical tools and the proper mathematical setup, we end up with meaningful answers, no matter what.

Question 3

For which of the following instances is liquidity most important?

- If the portfolio contains positions in T-notes or T-bonds
- Rebalancing portfolios
- Passive (static) asset allocation
- Estimating the parameters contained in $M_1, M_2, M_3,$ and M_4

Question 4

We have one foreign investment and hedge it with forward contracts. The volatility of the asset is $\sigma_a = 15\%,$ and the volatility of the forward contract is $\sigma_f = 11\%.$ In addition, the correlation between the asset and the forward contract is $\rho(a, f) = 0.4$.

According to the lecture notes, what is the optimal number of forward contracts?

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- -1
- 0.545
- -0.293
- -0.545

Question 5

We have available the returns for a manager's portfolio over a period of T months. For the same period, we also have the returns for four indexes: small-cap growth (SCG), large-cap growth (LCG), small-cap value (SCV), and large-cap value (LCV).

We estimate the following linear regression model:

$$\widehat{R}_{pt} = 0.01 + 0.2 \cdot SCG_t + 0.01 \cdot LCG_t + 0.5 \cdot SCV_t + 0.29 \cdot LCV_t$$

The manager claims that her investment style is to invest mostly in large-cap and value stocks.

What can we say about her claims?

- Correct about the value stock style, correct about the large-cap style.
- Incorrect about the value stock style, correct about the large-cap style.
- Incorrect about both the value stock and the large-cap styles.
- Correct about the value stock style, incorrect about the large cap.

Question 6

Sabine Schwinsteiger, a German portfolio manager is using Michaud's resampled efficient frontiers model for which she has obtained a license to construct a portfolio for her clients.

Which of the following statements is **false**?

- Sabine is not allowed to use Michaud's technique as it is patented. She should use Morningstar's instead.
- Sabine can use Michaud's methodology for she has purchased a license to do so.
- Sabine is allowed to use Michaud's methodology, but that does not always mean that the portfolio she identifies for her clients is without any issue.
- Sabine will report to her clients that projections and estimates were obtained on a best effort basis and cannot be guaranteed to show the actual effective outcome of investing in that portfolio.

Question 7

Describe what the goal of this portfolio optimization is.

$$\max_{w} \alpha_1 w^T M_1 + \alpha_2 w^T M_2 w + \alpha_3 w^T M_3 (w \otimes w) + \alpha_4 w^T M_4 (w \otimes w \otimes w)$$
$$\text{subject to} \quad w^T \iota = 1$$
$$w^T M_3 (w \otimes w) \geq -0.5$$

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- This is a portfolio optimization problem using mean, variance, skewness, and kurtosis. The constraints are
  1. no short sales allowed, and
  2. skewness must not be less than -0.5.
- This is a classic mean-variance portfolio optimization problem. The constraints are
  1. weights sum to 1, and
  2. portfolio's skewness must be greater than or equal to -0.5.
- The objective function uses mean, variance, skewness, and kurtosis, and the constraints require that
  1. the weights sum to 1, and
  2. the skewness of the portfolio is greater than or equal to -0.5.
- (●) This is a classic mean-variance portfolio optimization problem. The constraints are
  1. weights sum to 1, and
  2. portfolio's excess kurtosis must be greater than or equal to -0.5.

Question 8

We have seen that when dealing with multiple currencies, the mean-variance optimal portfolio can be obtained by solving the following optimization program:

$$\begin{aligned} &\max_{w} w^T r^{(d)} - \frac{1}{2} w^T \Sigma^{(d)} w \\ &\text{subject to } \; w^T \iota = 1 \end{aligned}$$

How would the program change, if all exposures to foreign currencies were fully hedged?

- $$\begin{aligned} &\max_{w} w^T r^{(d)} - \frac{1}{2} w^T \Sigma^{(f)} w \\ &\text{subject to } \; w^T \iota = 1 \end{aligned}$$
- We would still solve the program:
  $$\begin{aligned} &\max_{w} w^T r^{(d)} - \frac{1}{2} w^T \Sigma^{(d)} w \\ &\text{subject to } \; w^T \iota = 1 \end{aligned}$$
- $$\begin{aligned} &\max_{w} w^T r^{(f)} - \frac{1}{2} w^T \Sigma^{(d)} w \\ &\text{subject to } \; w^T \iota = 1 \end{aligned}$$
- (●) $$\begin{aligned} &\max_{w} w^T r^{(f)} - \frac{1}{2} w^T \Sigma^{(f)} w \\ &\text{subject to } \; w^T \iota = 1 \end{aligned}$$

Question 9

Which of the factors are mostly not observable?

- Fundamental factors
- Statistical factors
- Macroeconomic factors

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- Industry factors

Question 10

Which of the following optimization problems is **not** legitimate for computing an optimal portfolio?

- $\max_w w^T r - \frac{1}{2} w^T \Sigma w$
  $\text{s.t.} \quad w^T \iota = 1,$
  $w_1 > 0, w_2 < 0.3$
- $\max_w w^T r - \frac{1}{2} w^T \Sigma w$
  $\text{s.t.} \quad w^T \iota = 1$
- $\min_w \frac{1}{2} w^T \Sigma w - w^T r$
  $\text{s.t.} \quad w^T \iota = 1$
- $\max_w -w^T r - \frac{1}{2} w^T \Sigma w$
  $\text{s.t.} \quad w^T \iota = 1$

Question 11

Assume that W, Z, Y, Z are independent and identically distributed random variables with mean μ and variance σ^2. Compute the K_{WXZY} element of the cokurtosis matrix.

- 4
- 0
- -1
- 1

Question 12

Suppose we have stocks in three different currencies, A, B, and C. One stock is denominated in currency A, one in currency B, and three in currency C. What is the Q matrix in this case?

- $Q = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$
- $Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$
- $Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

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◉ $Q = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

Question 13

If we have 6 assets and 3 factors over a period of $T = 50 weeks$, what is the dimension of the matrix that contains the loading coefficients (assuming coefficients are fixed across periods) when considering the cross-section regression model?

- $3 \times 6.$
- $3 \times 1.$
- $6 \times 3.$
- ◉ $300 \times 3.$

Question 14

Which of the statements below is correct?

- Michaud's efficient resample method, like most resampling methods, is just wishful thinking and does not truly provide superior results, just nice plots to look at.
- Michaud's efficient resample method is robust, and we would not really benefit from the use of shrinkage estimators.
- Michaud's efficient resample method only benefits from shrinkage estimators for the covariance matrix as there are many more parameters than for the mean returns to estimate.
- ◉ Michaud's efficient resample method can be improved by using shrinkage estimators for mean returns and variance-covariance matrix of returns.

Question 15

Consider the following general constrained optimization problem:

$$\begin{aligned} \max_{w_1,w_2,w_3,w_4} \quad & w_1^2 + w_1 w_2 w_3^2 + w_4 e^{-w_4} \\ \text{subject to} \quad & w_1^2 + w_2^2 + w_3^2 + w_4^2 \geq 5 \\ & w_1 = 2w_2 \\ & 1 \leq w_i \leq 5, \; i = 1, 2, 3, 4 \end{aligned}$$

What is the optimal solution? (You will need to use Python to solve this problem and you can use the following initial conditions $w_2, w_2 = 3, w_3 = 2, w_4 = 1.$)

- Optimal value is 338.92
- Optimal value is -338.92
- Optimal value is 325.48.
- ◉ Optimal value is 337.87

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Question 16

Assume we have a position invested in a foreign currency denominated asset.

For each type of event, identify the proper action in regards to hedging currency exposure.

A) short strangle or straddle

B) carry trade

C) Currency volatility increases

D) reduce hedge

Foreign currency price appreciates	A
Market conditions are stable	B
Volatility of currency decreases	C

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